

Skill Week-10

Session-19

Task 1: To Register Handlers, Configure Signal Actions, Handle Interrupts, Manage Signal Context, Verify Stability, Test Signal Events.

```
nirvighna@DESKTOP-VS57D2I:~$ mkdir -p ~/OSSP/SignalJobs
nirvighna@DESKTOP-VS57D2I:~$ cd ~/OSSP/SignalJobs
nirvighna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ pwd
/home/lenovo/OSSP/SignalJobs
nirvighna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ nano signal_handler.c
nirvighna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ gcc signal_handler.c -o signal_handler -Wall -Wextra -g -Wall -Wextra
nirvighna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ ./signal_handler
=====
SIGNAL HANDLER PROGRAM
=====
Program started successfully.
PID = 1392

Press Ctrl+C to send SIGINT.
Use 'kill -TERM PID' to send SIGTERM.
=====
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
^C
Signal received!
Signal number: 2
Sender PID: 0
SIGINT received. Program will continue running.
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
```

```

GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <string.h>

volatile sig_atomic_t signal_received = 0;

void signal_handler(int sig, siginfo_t *info, void *context)
{
    // Prevent unused parameter warning
    (void)context;

    signal_received = sig;

    printf("\nSignal received!\n");
    printf("Signal number: %d\n", sig);
    printf("Sender PID: %d\n", info->si_pid);

    if (sig == SIGINT)
    {
        printf("SIGINT received. Program will continue running.\n");
    }
    else if (sig == SIGTERM)
    {
        printf("SIGTERM received. Program will terminate.\n");
        exit(0);
    }
}

int main()
{
    struct sigaction action;

    // Clear the structure
    memset(&action, 0, sizeof(action));

    // Set signal handler
    action.sa_sigaction = signal_handler;
    action.sa_flags = SA_SIGINFO;

    // Initialize signal mask
    sigemptyset(&action.sa_mask);

    // Register SIGINT handler
    if (sigaction(SIGINT, &action, NULL) == -1)
    {
        perror("sigaction SIGINT");
        return 1;
    }

    // Register SIGTERM handler

```

```

// Register SIGTERM handler
if (sigaction(SIGTERM, &action, NULL) == -1)
{
    perror("sigaction SIGTERM");
    return 1;
}

printf("=====\n");
printf("    SIGNAL HANDLER PROGRAM\n");
printf("=====\n");
printf("Program started successfully.\n");
printf("PID = %d\n", getpid());
printf("\nPress Ctrl+C to send SIGINT.\n");
printf("Use 'kill -TERM PID' to send SIGTERM.\n");
printf("=====\n");

// Keep program running
while (1)
{
    printf("Program is running...\n");
    sleep(2);
}

return 0;
}

```

Task 2: To Capture SIGINT, Forward Signals, Protect Shell Process, Terminate Foreground Jobs, Update Job States, Test Interrupt Handling.

```
nirvaghna@DESKTOP-VS57D2I:~$ cd ~/OSSP/SignalJobs
nirvaghna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ nano sigint_forward.c
nirvaghna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ gcc sigint_forward.c -o sigint_forward -Wall -Wextra -Wall -Wextra
nirvaghna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ ./sigint_forward
Parent: Foreground job PID = 2553
Press Ctrl+C to send SIGINT.
Foreground job started.
Child PID = 2553
Child: Working... 1
Child: Working... 2
Child: Working... 3
Child: Working... 4
Child: Working... 5
Child: Working... 6
Child: Working... 7
Child: Working... 8
^C
Parent: SIGINT received.
Child: Working... 9

Parent: SIGINT received.
Parent: Forwarding SIGINT to child PID 2553.
Parent: Foreground job exited normally.
Parent: Job state updated.
Parent: Shell-like process is still running.

Parent: SIGINT received.
Child: Working... 10
nirvaghna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ Child: Working... 11
Child: Working... 12
Child: Working... 13
Child: Working... 14
Child: Working... 15
Child: Working... 16
Child: Working... 17
Child: Working... 18
^C
nirvaghna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ Child: Working... 19
Child: Working... 20
Child: Finished normally.
```

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

pid_t foreground_pid = -1;

void sigint_handler(int sig)
{
    // Prevent unused parameter warning
    (void)sig;

    printf("\nParent: SIGINT received.\n");

    if (foreground_pid > 0)
    {
        printf("Parent: Forwarding SIGINT to child PID %d.\n",
            foreground_pid);

        kill(foreground_pid, SIGINT);
    }
}

int main()
{
    struct sigaction action;
    pid_t pid;
    int status;

    // Configure SIGINT handler
    action.sa_handler = sigint_handler;
    sigemptyset(&action.sa_mask);
    action.sa_flags = 0;

    if (sigaction(SIGINT, &action, NULL) == -1)
    {
        perror("sigaction");
        return 1;
    }

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
```

```

    // Child process
    printf("Foreground job started.\n");
    printf("Child PID = %d\n", getpid());

    for (int i = 1; i <= 20; i++)
    {
        printf("Child: Working... %d\n", i);
        sleep(1);
    }

    printf("Child: Finished normally.\n");
    exit(0);
}

// Parent process
foreground_pid = pid;

printf("Parent: Foreground job PID = %d\n", pid);
printf("Press Ctrl+C to send SIGINT.\n");

// Wait for child
waitpid(pid, &status, 0);

if (WIFEXITED(status))
{
    printf("Parent: Foreground job exited normally.\n");
}
else if (WIFSIGNALED(status))
{
    printf("Parent: Foreground job terminated by signal %d.\n",
           WTERMSIG(status));
}

foreground_pid = -1;

printf("Parent: Job state updated.\n");
printf("Parent: Shell-like process is still running.\n");

return 0;
}

```

Task 1: To Capture SIGTSTP, Suspend Processes, Update Job Table, Preserve State, Resume Later, Test Suspension Workflow.

```
nirvighna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ nano sigtstp_demo.c
nirvighna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ gcc sigtstp_demo.c -o sigtstp_demo -Wall -Wextra -lm -Wall -Wextra
nirvighna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$ ./sigtstp_demo
Parent: Child PID = 3560
Child started. PID = 3560
Child: Working 1/20
Press Ctrl+Z to suspend the child.
Child: Working 2/20
Child: Working 3/20
Child: Working 4/20
Child: Working 5/20
Child: Working 6/20
Child: Working 7/20
Child: Working 8/20
Child: Working 9/20
Child: Working 10/20
Child: Working 11/20
^Z
Parent: SIGTSTP received.
Parent: SIGTSTP received.
Child: Working 12/20
Parent: Stopping child PID 3560.
^Z
Parent: SIGTSTP received.
Parent: Stopping child PID 3560.
Parent: Child finished.
Parent: Job state updated to DONE.
nirvighna@DESKTOP-VS57D2I:~/OSSP/SignalJobs$
```

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

pid_t child_pid = -1;

void sigtstp_handler(int sig)
{
    // Prevent unused parameter warning
    (void)sig;

    printf("\nParent: SIGTSTP received.\n");
    if (child_pid > 0)
    {
        printf("Parent: Stopping child PID %d.\n", child_pid);
        kill(child_pid, SIGSTOP);
    }
}

int main()
{
    struct sigaction action;
    pid_t pid;
    int status;

    // Configure SIGTSTP handler
    action.sa_handler = sigtstp_handler;
    sigemptyset(&action.sa_mask);
    action.sa_flags = 0;

    if (sigaction(SIGTSTP, &action, NULL) == -1)
    {
        perror("sigaction");
        return 1;
    }

    pid = fork();
    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
        printf("Child started. PID = %d\n", getpid());
    }
}
```

```

// Child process
printf("Child started. PID = %d\n", getpid());
for (int i = 1; i <= 20; i++)
{
    printf("Child: Working %d/20\n", i);
    sleep(1);
}
printf("Child: Finished.\n");
exit(0);
}

// Parent process
child_pid = pid;
printf("Parent: Child PID = %d\n", child_pid);
printf("Press Ctrl+Z to suspend the child.\n");

// Wait until child is stopped
waitpid(child_pid, &status, WUNTRACED);
if (WIFSTOPPED(status))
{
    printf("\nParent: Child is now STOPPED.\n");
    printf("Parent: Job state updated to STOPPED.\n");
    printf("Parent: Sending SIGCONT...\n");
    // Resume child
    kill(child_pid, SIGCONT);
    printf("Parent: Child resumed.\n");
}

// Wait until child finishes
waitpid(child_pid, &status, 0);
if (WIFEXITED(status))
{
    printf("Parent: Child finished.\n");
    printf("Parent: Job state updated to DONE.\n");
}

return 0;
}

```

Task 2: To Create Process Groups, Assign Group IDs, Transfer Terminal Control, Restore Ownership, Coordinate Signals, Validate Behavior.

```

nirvighna@DESKTOP-VS57D2I:~$ nano process_group.c
nirvighna@DESKTOP-VS57D2I:~$ gcc process_group.c -o process_group -Wall -Wextra up -Wall -Wextra
nirvighna@DESKTOP-VS57D2I:~$ ./process_group
Child 1: PID = 4242
Child 1: PGID = 4242
Child 1: Working 1/5

Parent PID = 4241
Process Group ID = 4242
Transferring terminal control to process group 4242...
Foreground process group is now 4242
Child 2: PID = 4243
Child 2: PGID = 4242
Child 2: Working 1/5
Child 1: Working 2/5
Child 2: Working 2/5
Child 2: Working 3/5
Child 1: Working 3/5
Child 2: Working 4/5
Child 1: Working 4/5
Child 2: Working 5/5
Child 1: Working 5/5

Restoring terminal control to parent...

[1]+  Stopped                  ./process_group
nirvighna@DESKTOP-VS57D2I:~$

```

```

GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>
#include <termios.h>

int main()
{
    pid_t child1;
    pid_t child2;
    pid_t pgid;
    int status;

    child1 = fork();

    if (child1 < 0)
    {
        perror("fork");
        return 1;
    }

    if (child1 == 0)
    {
        // First child becomes process-group leader
        setpgid(0, 0);

        printf("Child 1: PID = %d\n", getpid());
        printf("Child 1: PGID = %d\n", getpgrp());

        for (int i = 1; i <= 5; i++)
        {
            printf("Child 1: Working %d/5\n", i);
            sleep(1);
        }

        exit(0);
    }

    // Parent ensures child 1 is group leader
    setpgid(child1, child1);

    pgid = child1;

    child2 = fork();

    if (child2 < 0)
    {
        perror("fork");
        return 1;
    }

```

```

    if (child2 == 0)
    {
        // Put second child in child 1's group
        setpgid(0, pgid);

        printf("Child 2: PID = %d\n", getpid());
        printf("Child 2: PGID = %d\n", getpgrp());

        for (int i = 1; i <= 5; i++)
        {
            printf("Child 2: Working %d/5\n", i);
            sleep(1);
        }

        exit(0);
    }

    // Parent places child 2 into the same group
    setpgid(child2, pgid);

    printf("\nParent PID = %d\n", getpid());
    printf("Process Group ID = %d\n", pgid);

    printf("Transferring terminal control to process group %d...\n",
        pgid);

    if (tcsetpgrp(STDIN_FILENO, pgid) == -1)
    {
        perror("tcsetpgrp");
    }

    printf("Foreground process group is now %d\n", pgid);

    // Wait for both children
    waitpid(child1, &status, 0);
    waitpid(child2, &status, 0);

    // Restore terminal control
    printf("\nRestoring terminal control to parent...\n");

    if (tcsetpgrp(STDIN_FILENO, getpgrp()) == -1)
    {
        perror("tcsetpgrp");
    }

    printf("Terminal control restored.\n");
    printf("All child processes completed.\n");

    return 0;
}

```

